

Communication

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	Slack / Discord	Core Developers & ML Engineers	Track daily micro-tasks, local dataset pipeline states, and algorithm blockers.
2	Progress Update	Weekly	GitHub Projects / Notion	Development Team & Mentor	Evaluate milestone status on model training convergence and web framework routing.
3	Issue / Bug Discussion	As Needed	WhatsApp / GitHub Issues	Backend Engineers & QA	Debug database API latency issues, batch ingestion exceptions, or prediction mismatches.
4	Stakeholder Review	Bi-Weekly	Google Meet	Team Lead, Mentor, Evaluator	Demonstrate dynamic analytics dashboards, yield forecasts, and live engine capabilities.
5	Final Demo Rehearsal	Once	Google Meet	All Team Members	Execute full walk-through dry run of the live cloud deployment and system execution flow.

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6	Data Pipeline Review	As Needed	Miro / Jupyter Labs	Data Scientists & Core Devs	Align data transformations, historical feature mapping, and feature ingestion metrics.

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Asynchronous schedule conflicts during initial system configuration stages made tracking machine learning features tricky.	Migrated tracking workflows to a unified dashboard in GitHub Projects and tracked data schema changes directly in shared repository logs.
2	High testing query traffic caused occasional data sync limits to overflow when simulating simultaneous regional telemetry requests.	Implemented a local data-caching layout inside the testing environment and updated the system architecture to use local queue listeners.
3	Ambiguities regarding the required format of structured API payload parameters between front-end visuals and backend ML libraries.	Scheduled immediate peer code reviews to finalize cross-component endpoints and updated the project integration specifications manual.